

DM SG 3 – Anatomy - Gastrointestinal System

Dress code: Full Scrubs and Closed Toe Shoes

Required materials and equipment:

1. Stethoscope
2. A copy of the clinical cases
3. A copy of the physical examination of the abdomen handout
4. An electronic copy or hard copy of your Netter Atlas(required) and lecture notes(optional)
5. Turning point clicker device

How to prepare for and what to expect from this session

1. Research Article: 15 minutes

Use library resources or open resources (eg. Google Scholar) to find a peer reviewed article on Meckel's Diverticulum. Use the following objectives as a guide to review the selected paper. Once you are able to answer most of these questions, then you have successfully completed your review and are fully prepared for discussion.

- a) How did you find the article?
- b) Why did you choose the specific article?
- c) Outline the basic embryological origin of the diverticulum.
- d) Describe Meckel's diverticulum in terms of the rule of two's.
- e) What are the alternative names for the omphalomesenteric duct?
- f) Describe the ectopic tissues that may be found in Meckel's diverticulum.
- g) Describe some variations in the location of Meckel's diverticulum.
- h) Describe the basic clinical features of Meckel's diverticulum.

2. Clinical Case Discussion: 45 minutes

Use your lectures, DLAs and textbooks to answer the clinical case questions listed below (Cases A-C). These questions will be discussed with your peers and clinical facilitators during the small group. **At the end of the session, all students are responsible to have completed all the learning objectives listed for that session.** If a group is not able to finish all learning objectives in allotted time, students have the individual responsibility for learning them. If a group finishes all learning objectives before the allotted time has ended, the students must reinforce the learned concepts by inviting interactions with Facilitators, peers and subject experts.

Case A

Objectives:

- SOM.MKII.BPM2.1.DM.1.ANAT.1185 Describe the anatomy of the stomach and identify its parts and curvatures.
- SOM.MKII.BPM2.1.DM.1.ANAT.1186 Describe the relationships of the stomach to the spleen, pancreas, liver, ascending and descending colon.
- SOM.MKII.BPM2.1.DM.1.ANAT.1187 Describe the blood supply and the venous and lymphatic drainage of the stomach.
- SOM.MKII.BPM2.1.DM.1.ANAT.1188 Compare and contrast where stomach contents from a perforation of the posterior, versus the anterior stomach wall will initially accumulate.
- SOM.MKII.BPM2.1.DM.1.ANAT.1189 List the blood vessels which may be eroded in ulceration of the posterior wall of the stomach.

A 48-year-old man was diagnosed with stomach cancer. During surgery for partial gastrectomy, it was noticed that the posterior wall of the stomach was adherent to some structures. The surgeon proceeded with great caution and isolated the stomach without injuring the structures behind it.

1. What structures form the stomach bed?
2. What arteries supply the different areas of the stomach?
3. What is the embryological origin of the stomach [epithelium, musculature & nerve plexus]?
4. What artery is at risk if this cancer progresses through the posterior wall of the stomach?
5. Into which part of the peritoneal cavity will the contents of the stomach spill if perforation might occur?
6. What autonomic nerve component regulates gastric acid secretion and motility?

Case B

Objectives:

- SOM.MKII.BPM2.1.DM.1.ANAT.1193 Describe the different portions of the small intestine and explain the external and internal differences between these parts.
- SOM.MKII.BPM2.1.DM.1.ANAT.1194 Describe the location of the duodenum and its relationship to surrounding organs.
- SOM.MKII.BPM2.1.DM.1.ANAT.1195 Describe the blood supply and the venous and lymphatic drainage of the duodenum, jejunum and ileum.
- SOM.MKII.BPM2.1.DM.1.ANAT.1196 Describe the sympathetic, parasympathetic and sensory innervation of the foregut, midgut and hind gut.
- SOM.MKII.BPM2.1.DM.1.ANAT.1197 Describe the location of the referred pain from the gastrointestinal tract and organs based on innervation and embryological origin.

A 24-year-old medical student awoke one morning feeling somewhat nauseated. He then felt generalized pain around his umbilical region. Some hours later, the pain became sharp and localized to the right lower quadrant. At the hospital, during palpation of his abdomen, there was an increase in the pain over his right lower quadrant at McBurney's point. Further examination led to the diagnosis of appendicitis, following which the appendix was surgically removed.

1. Give the anatomical explanation for the pain felt over the umbilical region.
2. Explain what caused the pain to be sharp and localized in the area of the right lower quadrant and why this pain increased on palpation.
3. Explain what is McBurney's point its clinical significance.
4. What anatomical landmark could be used by the surgeon to locate the appendix upon opening the peritoneal cavity?
5. What artery supplies the appendix and where does this artery usually originate?

Case C

Objectives:

- **SOM.MKII.BPM2.1.DM.1.ANAT.1205 Describe the anatomy of the biliary tree.**
- **SOM.MKII.BPM2.1.DM.1.ANAT.1206 Describe the location of the pancreas and its relationship to the surrounding structures.**
- **SOM.MKII.BPM2.1.DM.1.ANAT.1207 Describe the blood supply, venous and lymphatic drainage of the pancreas.**
- **SOM.MKII.BPM2.1.DM.1.ANAT.1208 Describe how pancreatic enzymes flow to the duodenum.**

A 65-year-old man was admitted to the hospital for weight loss and a lack of energy and interest in doing things (lethargy). On physical examination, the physician noted that the patient was emaciated and there was yellowing of the skin and eyes. A well-defined rounded mass was palpable below the liver edge in the right upper quadrant. Endoscopic retrograde pancreatography was performed revealing a large mass in the head of the pancreas. The histopathological report of a biopsy of the mass indicated it was a carcinoma.

1. Name the parts of the biliary tree
2. What is the cause of jaundice in this patient?
3. What are the structures that can be compressed by the growth?
4. Describe the termination of the major pancreatic duct?
5. What caused the palpable mass in the abdomen?
6. What major artery passes between the head & uncinate process of the pancreas?
7. What are the origins of the arteries that supply the head of pancreas?

3. Physical Examination Video: 40 minutes

- a) Review the abdominal examination skills practice handout
- b) Skills Checklist: Use the following checklist as a guide to the physical examination techniques that you are responsible for knowing by the end of the session.

Position the patient and expose the area to be examined appropriately
Inspection
Symmetry
Scars, striae, rashes, lesions or dilated veins
Contour: protuberant, flat, scaphoid
Visible peristalsis
Aortic pulsations
Auscultation
Bowel sounds
Bruits over aorta, renal arteries, common iliac and femoral arteries
Percussion
Percuss in all 4 quadrants and/or 9 regions with proper technique
Palpation
Light palpation for tenderness, guarding, or superficial masses
Deep palpation for tenderness and masses
Palpate for rebound tenderness
Examination of the Liver
Percuss liver span
Palpate liver edge
Examination of the Spleen
Palpate spleen
Percuss spleen
Examination of the kidney
Ballottement of the kidney

4. Small Group Evaluation: 5 minutes

- a) Facilitator and small group evaluations